

llmXive Automated Review of NatureBench: Can Coding Agents Match the Published SOTA of Nature-Family Papers?

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_claim_accuracy.md

The paper presents a rigorous benchmarking framework, but several central claims rely on model names and statistics that do not align with current public reality or internal data consistency.

First, the evaluation section (Section 4.1) and Table 1 list “GPT-5.5”, “Gemini 3.5 Flash”, and “Claude Opus 4.7” as evaluated agents. These model versions do not exist in the public record as of this review. Citing non-existent baselines for a “SOTA” comparison invalidates the core empirical claim that agents “surpass SOTA on only 17.8% of tasks,” as the “SOTA” itself is undefined or hypothetical. The authors must verify the actual model versions used (e.g., GPT-4o, Claude 3.5 Sonnet) and update the manuscript to reflect reproducible, real-world baselines.

Second, there is a numerical inconsistency in the analysis. Section 5.1 claims the aggregate “Match-SOTA rate of all agents is only 32.2%”. A simple arithmetic mean of the “Match-SOTA” column in Table 1 yields approximately 31.2%. If the 32.2% figure represents a weighted average (e.g., by task difficulty or instance count), this methodology is not stated. Without clarification, this specific statistic appears unsupported by the provided table data.

Finally, the bibliography references (e.g., anthropic2026opus47) point to future-dated or non-existent papers. While the paper is a preprint, the scientific claim depends on the existence of these models. If these are placeholders for future work, the claims must be qualified; if they are errors, the text must be corrected to match the actual experimental setup.

Action items.

- **[writing]** The paper presents a rigorous benchmarking framework, but several central claims rely on model names and statistics that do not align with current public reality or internal data consistency. First, the evaluation section (Section 4.1) and Table 1 list “GPT-5.5”, “Gemini 3.5 Flash”, and “Claude Opus 4.7” as evaluated agents. These model versions do not exist in the public record as of this review. Citing non-existent baselines for a “SOTA” comparison invalidates the core empirical claim that age

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 is a clear and well-structured workflow diagram that effectively visualizes the NatureGym pipeline described in the caption. All stages, inputs, outputs, and the information firewall concept are explicitly labeled and logically connected.

Figure 2

The figure effectively summarizes the corpus composition with clear domain and task family breakdowns, but the ‘ML task family’ chart includes a legend entry without a corresponding bar, and the ‘Contribution type’ chart lacks a legend to interpret its symbols.

Figure 3

The figure effectively visualizes the calibration probe and outcomes, but the caption for panel c contains a rendering error (‘\$\$’), and panel c lacks its own legend for the model symbols.

Figure 4

The figure effectively visualizes the NatureBench pipeline and evaluation protocol with clear separation of agent and host components. However, the ‘Source Journals’ chart lacks axis labels for its values, and the bottom section detailing the agent’s iterative process is omitted from the caption description.

Figure 5

Figure 5 effectively communicates the distribution of agent performance relative to SOTA, but the x-axis scale is asymmetric (0-80 vs 0-40), which visually minimizes the ‘Match or exceed SOTA’ results. Additionally, the y-axis labels are cluttered, reducing readability.

Figure 6

Figure 6 effectively summarizes solution mechanisms and failure modes with clear bar charts and data labels. However, the x-axis label in panel (b) is ambiguous regarding whether it shows a rate or a percentage of total runs, and panel (c) contains a minor grammatical inconsistency in its y-axis label.

Figure 7

Figure 7 effectively visualizes performance across domains and task types, but Figure 7c contains a rendering error in the y-axis label (‘Spearman \$\$’) and has inconsistent label orientation.

Figure 8

Figure 8 presents agent trajectories but suffers from missing axis labels in panel (b) and undefined x-axis categories in panel (c), which obscures the interpretation of the decoding

strategy comparison.

Figure 9

The figure is visually clear with appropriate axes and labels, but the item counts in the caption (eight networks, 19 sub-tasks) contradict the number of bars actually rendered in panels (a) and (b).

Figure 10

Figure 10 provides a high-level overview of the domains and leaderboard, but the specific citations listed in the caption for panel (a) are not visible in the image, and the y-axis label in panel (b) lacks the explicit threshold definition found in the caption.

Action items.

- **[writing]** Figure 2: The ‘ML task family’ bar chart contains a legend entry ‘Tail 19’ that is not represented by a visible bar in the chart, creating a mismatch between the legend and the data visualization.
- **[writing]** Figure 2: The ‘Contribution type’ bar chart uses colored dots to represent categories (Adapt, Imax, Form) but lacks a legend or key to define what these colors or symbols signify.
- **[writing]** Figure 3c: The caption text ‘Per-task

inbaseversusreproducemode’containsarenderingartifact(

’) instead of the variable name (likely ‘g’), making the description unreadable.

- **[science]** Figure 3c: The scatter plot lacks a legend defining the blue circles and red triangles, which are only identified by model names in the legend of Figure 3b; this forces the reader to cross-reference panels to understand the data series.
- **[writing]** Figure 4: The ‘Source Journals’ bar chart in the top-left panel lacks a visible x-axis or unit labels, making the numerical values (36, 26, 16, etc.) ambiguous without external context.
- **[writing]** Figure 4: The ‘Agent Iterative Process’ section at the bottom is not mentioned in the caption, which focuses on the pipeline construction and evaluation service rather than the agent’s internal method evolution.
- **[science]** Figure 5a: The x-axis scale is misleading and inconsistent. The left side (negative gap) spans 0 to 80, while the right side (positive gap) spans 0 to 40. This compresses the visual representation of the ‘Match or exceed SOTA’ performance, making the blue bars appear smaller than they are relative to the red bars, which distorts the comparison of success vs. failure rates.
- **[writing]** Figure 5a: The y-axis labels are cluttered and difficult to read. Model names (e.g., ‘Claude Opus 4.7’) are stacked with sub-labels (e.g., ‘Claude Code’) in a way that creates visual noise and makes it hard to associate the correct bar with the correct model.

- **[science]** Figure 6b: The x-axis label ‘% of runs / Match-SOTA rate’ is ambiguous and potentially misleading. The bars represent the percentage of runs within specific method-family categories (Same family vs. Alternative) that achieved Match-SOTA, not a rate normalized by the total number of runs. The label should clarify that these are conditional success rates (e.g., ‘Match-SOTA rate within method family’) to avoid confusion with the overall run distribution shown in Figure 6a.
- **[writing]** Figure 6c: The y-axis label ‘Proxy prediction’ is likely a misnomer or typo for ‘Proxy-based prediction’ or ‘Proxy prediction method’ to match the grammatical structure of the other categories (e.g., ‘Search / tuning’, ‘Engineering pipeline’). While understandable, the phrasing is inconsistent with the other labels.
- **[writing]** Figure 7c: The y-axis label contains a LaTeX artifact ‘Spearman \$\$’ instead of the variable name (likely \$ho\$), and the axis lacks a title.
- **[writing]** Figure 7c: The y-axis label ‘Opus 4.7’ is rotated 90 degrees, making it difficult to read compared to the horizontal labels in other subplots.
- **[writing]** Figure 7c: The x-axis label ‘Spearman *ho*’ is present, but the y-axis label is missing the variable name, showing only ‘Spearman \$\$’.
- **[science]** Figure 8b: The y-axis lacks a label and units, making the ‘Relative gap’ scale ambiguous for the genomic sequence prediction task.
- **[science]** Figure 8c: The x-axis labels (‘Greedy’, ‘Beam’) are not defined in the caption or legend, leaving the comparison between decoding strategies unclear.
- **[writing]** Figure 8: The green text ‘crosses SOTA’ in panel (a) is not defined in the caption or legend, making its meaning ambiguous.
- **[writing]** Figure 9: The caption states panel (a) shows ‘eight biological networks’, but the rendered image only displays seven bars (MTG, LTG, PCNet, Multinet, IRefindex v9, STRINGdb, CPDB, IRefindex v15).
- **[writing]** Figure 9: The caption states panel (b) shows ‘19 genomic sub-tasks’, but the rendered image only displays 18 bars (Enhancers through H3K27ac).
- **[writing]** Figure 10: The caption lists six specific citations (e.g., miao2025multigate) for the source figures in panel (a), but the rendered image does not display these citations or labels, making it impossible to verify the mapping between the representative figures and the listed papers.
- **[writing]** Figure 10: The bar chart in panel (b) displays ‘Surpass-SOTA (%)’ on the y-axis, but the caption defines the metric as ‘Surpass-SOTA ($g > 0.1$)’. The figure should explicitly clarify if the percentage represents the proportion of tasks where $g > 0.1$ or if the threshold differs from the caption’s definition.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The paper is generally well-structured for a technical audience, but it relies on several acronyms and notation conventions that are introduced without explicit definition at their first point of use, which may stall a competent reader from an adjacent field (e.g., a computer vision researcher reading a systems paper, or vice versa).

The most significant omissions are the acronyms “SOTA” (State-of-the-Art), “SR” (Score Rate), and “CR” (Completion Rate). While “SOTA” is ubiquitous in ML, the paper uses it in the Introduction before ever spelling it out. Similarly, “SR” and “CR” appear in the analysis of results (Section 4.2) before being defined in the table captions or the main text. An adjacent-field reader might reasonably guess their meaning, but the lack of explicit definition forces them to infer rather than read, violating the self-contained requirement.

Additionally, the notation for the relative gap metric is slightly opaque. The LaTeX command `\gimp` is defined in the preamble, but the prose introduces the concept of the “relative gap” and then immediately uses the symbol g (rendered by `\gimp`) in Equation 1 without explicitly stating “Let g denote the relative gap.” While the equation context makes it clear, a strict reading suggests the symbol itself was not formally introduced in the text prior to the equation.

Finally, the “Tier S” classification for dataset sizes is mentioned in Section 2 and 3, but the specific size thresholds defining the tiers are not clearly summarized in a single location, requiring the reader to piece together the definition from scattered parentheticals.

These are minor fixes (adding a parenthetical or a defining clause) that would significantly improve accessibility without altering the scientific content.

Action items.

- **[writing]** The paper is generally well-structured for a technical audience, but it relies on several acronyms and notation conventions that are introduced without explicit definition at their first point of use, which may stall a competent reader from an adjacent field (e.g., a computer vision researcher reading a systems paper, or vice versa). The most significant omissions are the acronyms “SOTA” (State-of-the-Art), “SR” (Score Rate), and “CR” (Completion Rate). While “SOTA” is ubiquitous in ML, the pape

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper’s argument structure is generally sound, with clear premises leading to conclusions about agent capabilities. However, there are minor inconsistencies in the reporting of aggregate statistics that require clarification to ensure the logical flow from data to conclusion is unambiguous.

In Section 5.1, the text states: “As shown in Fig.~\ref{fig:solution-mechanisms}a, the Match-SOTA rate of all agents is only 32.2%“. This phrasing is ambiguous. Does it mean the average Match-SOTA rate across the 10 agents is 32.2%, or that 32.2% of the total 900 runs achieved Match-SOTA? Looking at Table 1 (main-results), the Match-SOTA rates for the 10 agents are: 47.8, 37.8, 44.4, 36.7, 28.9, 30.0, 27.8, 28.9, 26.7, 13.3. The arithmetic mean of these values is approximately 32.23%. Thus, the text likely refers to the mean across agents. However, the phrase “Match-SOTA rate of all agents” could be misinterpreted as a global rate (total successes / total runs). Given that the section discusses “900 task-agent runs,” the distinction is critical for the logical interpretation of the results. The text should be rephrased to “the mean Match-SOTA rate across agents is 32.2%” or “32.2% of the 900 runs achieved Match-SOTA” (if that is the intended meaning, though the math suggests the former).

Additionally, Section 4.2 states: “Match-SOTA rates ($g \geq 0$) are higher but still below half: Claude Opus 4.7 leads at 47.8%, followed by GPT-5.5 (44.4%), Gemini 3.5 Flash (37.8%), and Claude Opus 4.6 (36.7%). The remaining agents cluster between 26.7% and 30.0%“. This description is consistent with Table 1. The discrepancy arises only in Section 5.1 where the aggregate figure is presented without specifying it is a mean. This is a minor logical gap in clarity rather than a fatal contradiction, but it should be fixed to ensure the reader can perfectly trace the argument from the table to the text.

No other logical inconsistencies, contradictions between sections, or non-sequiturs were found. The definitions of metrics (g , Match-SOTA, Surpass-SOTA) are consistent throughout. The causal claims (e.g., failures dominated by wrong method choice) are supported by the ablation/annotation data presented in Section 5.1 and Figure 5.1d. The argument that agents succeed via “methodological translation” rather than “scientific invention” follows logically from the breakdown of success modes in Figure 5.1c.

The paper is logically consistent, but the statistical reporting in Section 5.1 needs a small clarification to avoid ambiguity.

Action items.

- **[writing]** The paper’s argument structure is generally sound, with clear premises leading to conclusions about agent capabilities. However, there are minor inconsistencies in the reporting of aggregate statistics that require clarification to ensure the logical flow from data to conclusion is unambiguous. In Section 5.1, the text states: “As shown in Fig.~\ref{fig:solution-mechanisms}a, the Match-SOTA rate of all agents is only 32.2%“. This phrasing is ambiguous. Does it mean the average Match-SOTA rate

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper presents a rigorous benchmarking framework, but several central claims in the abstract and conclusion slightly overreach the specific definitions and measurements provided in the results section.

First, the claim of being “the first” to combine paper-sourced tasks, scientific domains, and optimization-oriented evaluation (Abstract; Introduction) is not fully supported by Table 1. The table lists MLS-Bench, which evaluates “method invention” on “ML research problems” using “human baselines.” While NatureBench is distinct in using *Nature-family* papers and *published SOTA* anchors, the broader claim of being the first to combine these three axes ignores the existence of MLS-Bench. The claim should be narrowed to specify the unique contribution: e.g., “the first to evaluate discovery on tasks distilled specifically from Nature-family papers using published SOTA as the anchor.”

Second, the abstract and conclusion draw a sharp binary between “methodological translation” and “genuine scientific invention.” Section 5.1 quantifies success modes as “supervised proxy prediction” (45.5%) versus “domain-reasoned alternatives” (8.3%) and “method-aligned solutions” (9.0%). The paper does not explicitly define or measure “genuine scientific invention” as a separate category from “method-aligned” work. By framing the results as a failure of “invention” versus a success of “translation,” the rhetoric implies a capability gap that the data does not explicitly isolate. The text should be rephrased to reflect the measured categories: “agents succeed primarily by reducing scientific tasks to standard ML pipelines rather than by proposing novel domain-specific methods.”

Finally, the conclusion states that failures are “not by task misunderstanding,” citing the 3.1% “Understanding-layer” failure rate. However, the dominant “Method-layer” failures (61.1%) include “wrong method choice,” which can sometimes stem from a misunderstanding of the task’s scientific constraints. The absolute exclusion of task misunderstanding as a significant factor is slightly too strong given that the “Method-layer” category is broad. A more precise phrasing would be: “Failures are dominated by method selection and execution depth, with task misunderstanding accounting for a small fraction of cases.”

These are primarily rhetorical refinements to ensure the scope of the claims matches the granularity of the evidence presented.

Action items.

- **[writing]** Abstract/Intro claim NatureBench is ‘the first’ to combine paper-sourced tasks, science, and optimization. Table 1 shows MLS-Bench also targets ‘method invention’ on research problems. Narrow the claim to ‘first to use published SOTA anchors from Nature-family papers’ to match the evidence.
- **[writing]** Abstract contrasts ‘methodological translation’ with ‘genuine scientific invention,’

but Section 5.1 only measures ‘domain-reasoned alternatives’ (8.3%) and ‘method-aligned solutions’ (9.0%). Rephrase to ‘rather than proposing novel domain-specific methods’ to avoid implying a measured category that doesn’t exist.

- **[writing]** Conclusion states failures are ‘not by task misunderstanding’ (citing 3.1%). However, ‘wrong method choice’ (45.1%) can stem from misunderstanding. Soften to ‘Failures are dominated by method selection and execution depth, with task misunderstanding being a minor factor’ to match the data’s granularity.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

This paper presents a benchmark (NatureBench) and pipeline (NatureGym) for evaluating AI coding agents on scientific discovery tasks derived from Nature-family publications. From a safety and ethics perspective, the work is low-risk and appropriately scoped.

The primary data sources are publicly available datasets and code repositories associated with peer-reviewed papers, which the authors explicitly state are accessed without authentication or special application (Section 2.2, Level 3). The pipeline includes a “file-level firewall” (Section 2.3) and an “information firewall” (Section 2.1) designed specifically to prevent the leakage of source methods or ground truth to the agents, thereby mitigating risks of data contamination or shortcut-taking rather than genuine discovery.

The evaluation protocol involves running agents in isolated Docker containers with a strict “web-search-disabled” policy (Section 4.1), preventing the agents from accessing external sensitive information or the original papers during the task. The benchmark tasks themselves are derived from published scientific literature and do not involve generating novel biological sequences, chemical compounds, or cyber-attack vectors that could be directly weaponized; the “discovery” is limited to optimizing code for existing, public scientific problems.

There is no use of private human-subjects data, PII, or sensitive health records; the datasets are aggregated scientific data (e.g., genomic sequences, protein structures, molecular graphs) that are already public. The paper does not describe any dual-use capabilities that lower the barrier to harm (e.g., automated vulnerability discovery or biological synthesis planning) beyond the general capability of coding agents, which is the subject of the benchmark itself. No conflicts of interest are apparent that would bias the safety assessment, and the authors disclose their affiliations clearly.

Consequently, there are no foreseeable, non-trivial risks of harm that are unacknowledged or unmitigated in this work. The paper does not require additional safety disclosures or mitigations.

Scientific Evidence

Weighs the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents a rigorous benchmarking framework, but the evidentiary strength of its central quantitative claims is weakened by a lack of variance reporting and potential confounds in the experimental design.

First, the headline results in Table 1 (Section 4.2) are derived from single training/execution runs per agent-task pair. The paper reports that Claude Opus 4.7 surpasses SOTA on 17.8% of tasks, but provides no standard deviation, confidence interval, or seed count. In stochastic optimization and LLM-based agent workflows, performance can vary significantly across random seeds due to initialization, sampling, or non-deterministic training dynamics. A single run cannot distinguish a robust capability from a lucky initialization. To support the claim that these agents have a specific, stable capability profile, the authors must report results averaged over at least 3-5 seeds with mean $\pm$ standard deviation.

Second, the analysis of failure modes in Section 5.1 attributes 24.4% of failures to “insufficient compute budget.” This conclusion is drawn from a protocol where every task is strictly capped at 4 hours. The design confounds the agent’s inherent capability with the arbitrary time limit; it is impossible to know if an agent failed because it chose the wrong method or because it simply ran out of time to execute the correct one. To isolate the “compute” factor, the authors should run a compute-ablation study on a representative subset of tasks (e.g., extending the budget to 8 or 12 hours) to see if the failure rate drops or if the “wrong method” attribution changes.

Finally, the mechanistic claims regarding “methodological translation” (Section 5.1) rely on the manual annotation of 900 agent trajectories. The paper does not report the annotation rubric, the number of annotators, or the inter-annotator agreement (IAA) statistics. Without these, the precise percentages (e.g., 45.5% for translation) are subjective and lack the statistical rigor required to support such specific mechanistic conclusions. Reporting the IAA (e.g., Cohen’s kappa) and the detailed rubric is necessary to validate these qualitative-to-quantitative conversions.

Action items.

- **[writing]** The paper presents a rigorous benchmarking framework, but the evidentiary strength of its central quantitative claims is weakened by a lack of variance reporting and potential confounds in the experimental design. First, the headline results in Table 1 (Section 4.2) are derived from single training/execution runs per agent-task pair. The paper reports that Claude Opus 4.7 surpasses SOTA on 17.8% of tasks, but provides no standard deviation, confidence interval, or seed count. In stochastic optim

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The statistical treatment in this paper is generally sound in its descriptive reporting of the benchmark results, but it lacks inferential rigor in its comparative analysis sections.

Specifically, in Section 5.1 (“Solution Mechanisms”), the authors compare Match-SOTA rates between two groups of runs (same method family vs. different family): 37.7% vs. 29.6%. While the descriptive difference is clear, the text asserts this as a finding (“These shifts are not equally effective”) without reporting a statistical test (e.g., a chi-squared test or Fisher’s exact test) to determine if this difference is statistically significant. Given the sample size (900 runs), a test is feasible and necessary to support the claim of a systematic difference.

In Section 5.2 (“Domain and Interdisciplinary Variation”), the authors report Spearman correlations ($r = 0.71$) between agent rankings and the consensus domain ranking across six domains. With a sample size of only $n=6$ (the number of domains), a correlation of 0.71 is not statistically significant at the standard $\alpha=0.05$ level ($p = 0.12$). The claim that the ordering is “highly consistent” and “shared across agents” relies entirely on the magnitude of the correlation coefficient without acknowledging the low statistical power due to the small number of data points. The authors should either report the exact p-values to be transparent about the lack of significance or rephrase the claim to describe the observed trend without implying statistical significance.

Finally, while the benchmark results (Table 1) are reported with one decimal place (e.g., 17.8%), this precision suggests a level of certainty that may not exist if the agent runs are stochastic. The paper does not mention running agents with multiple random seeds to estimate the variance of these performance metrics. While single-run reporting is common in some agent benchmarks, the lack of any uncertainty measure (e.g., “based on a single run”) or variance reporting makes the point estimates appear more definitive than they are. A brief clarification on the single-run nature of the results or the inclusion of seed-based variance would improve the statistical transparency.

Action items.

- **[writing]** The statistical treatment in this paper is generally sound in its descriptive reporting of the benchmark results, but it lacks inferential rigor in its comparative analysis sections. Specifically, in Section 5.1 (“Solution Mechanisms”), the authors compare Match-SOTA rates between two groups of runs (same method family vs. different family): 37.7% vs. 29.6%. While the descriptive difference is clear, the text asserts this as a finding (“These shifts are not equally effective”) without reporting

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The paper is generally well-written, with a clear narrative arc and precise technical descriptions. The prose is dense but largely free of grammatical errors or confusing syntax. However, there are minor instances of structural redundancy and slightly clunky phrasing that, while not fatal, impede the smooth flow of reading.

In Section 2.1, the explanation of the pipeline stages is accurate but suffers from repetitive sentence structures (“First, it... Second, it...”). A more integrated sentence structure would improve readability without losing precision. Similarly, in Section 3.1, the transition between the journal selection and the pipeline funnel is slightly abrupt, with the sentence about the “other four” journals feeling like an afterthought rather than a seamless part of the narrative.

Section 4.2 repeats the “rarity” of improvements in the opening sentence of the paragraph, which was already established in the previous paragraph’s topic sentence. This redundancy slows the reader down. In Section 5.1, the breakdown of failure modes is clear, but the connection between the high-level summary and the specific percentages could be tightened to ensure the reader immediately grasps the relationship between the categories and the data.

Finally, the Appendix section titled “Authors” uses a non-standard header and formatting that diverges from the main body’s style, creating a minor visual and structural dissonance. While these issues do not obscure the scientific content, addressing them would elevate the paper from “clear” to “polished,” ensuring the reader moves through the argument with zero friction.

Action items.

- **[writing]** The paper is generally well-written, with a clear narrative arc and precise technical descriptions. The prose is dense but largely free of grammatical errors or confusing syntax. However, there are minor instances of structural redundancy and slightly clunky phrasing that, while not fatal, impede the smooth flow of reading. In Section 2.1, the explanation of the pipeline stages is accurate but suffers from repetitive sentence structures (“First, it... Second, it...”). A more integrated sentence